

Article

Coupled Electromagnetic–Fluid–Thermal Analysis of a Fully Air-Cooled Pumped Storage Generator Motor

Shukuan Zhang ^{1,2,3,*}, Fachen Wang ¹, Hongtao Wang ^{2,*}, Zhe Shao ¹, Hongge Zhao ¹ and Jingwei Zhu ¹ ¹ College of Marine Electrical Engineering, Dalian Maritime University, Dalian 116026, China² Motor Design and System Control of Fujian Provincial University Engineering Research Center (Ningde Normal University), Fujian Province University, Ningde 352100, China³ Key Laboratory of Special Machine and High Voltage Apparatus (Shenyang University of Technology), Ministry of Education, Shenyang 110870, China

* Correspondence: zhangshukuan@dmlu.edu.cn (S.Z.); t0937@ndnu.edu.cn (H.W.)

Abstract: With the continuous increase in the capacity of the pumped storage generator motor, the overheating of the rotor area is becoming increasingly severe, which has a significant effect on the safe and reliable operation of the machine. The heat dissipation of the machine rotor by fully air-cooled is one of the key technologies to develop the new generation of pumped storage generator motors. In this paper, the electromagnetic field and fluid–thermal coupled field of a pumped storage generator motor are analyzed. The 2D transient time-step finite element model of the electromagnetic field of a pumped storage generator motor is established, and the eddy current loss of damping bars of the rotor is calculated by the finite element method. The additional loss of the rotor pole surface is calculated by analytical method. The mathematical and geometric models of the 3D fluid–thermal coupled field of the pumped storage generator motor are established and calculated. The complex fluid velocity distribution and the temperature distribution at different positions of the rotor under fully air-cooled fanless cooling conditions are investigated in detail. The calculated temperature of field winding is compared with the measured value, and the result shows that the calculated result coincident well with the test data. This research provides the technical reference for the development and temperature rise calculation for large pumped storage generator motors.



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1. Introduction

With the rapid growth of the global economy, the continuous excessive consumption of non-renewable energy has resulted in a series of environmental problems. The world is actively promoting the green energy transformation and promoting the high-quality development of energy. Energy storage systems are vital electric grid facilities for promoting global clean energy transformation [1,2]. Pumped storage is the largest-scale grid-connected energy storage technology. A pumped storage unit can perform tasks such as peak shaving, valley filling, phase modulation, and frequency modulation in the power grid, greatly promoting the integration of wind and solar power, and ensuring the safe and reliable operation of the power system [3]. With the continuous development of pumped storage units, the power rating of pumped storage generator motors is increasing. The electromagnetic losses, field current, and temperature rise of the machine rotor increase obviously, which leads to overheating issues. Overheating of the machine rotor will seriously influence the safe and reliable operation of the pumped storage generator motor, which will cause instability of the power grid.

The fully air-cooled fanless cooling system of large hydrogenerators offers the benefits of robust reliability, low cost, and convenient operation and maintenance. However, the fully air-cooled fanless cooling system is an indirect cooling approach, and its cooling

effect is not as good as the direct cooling method, such as the water-cooled or hydrogen-cooled method, etc. Cooling air inside the machine is completely driven by the rotor's own rotation to take away the heat generation of the machine. Cooling air exhibits a complicated distribution within the alternator which is cooled by the fully air-cooled method. The temperature rise of various components of the machine is directly influenced by the distribution and flow rule of cooling air. Therefore, it is of great significance to study the fluid field distribution of cooling air and the thermal field of the machine rotor.

In recent years, some researchers have studied the cooling mechanism and temperature distribution of the large salient-pole synchronous alternator [4–13]. Bersch et al. developed a novel thermal and electromagnetic coupled model to obtain optimal ventilation design. To reduce the peak temperature of the stator winding, the radial ventilation of the stator was optimized and designed [14]. Han and Li et al. conducted a study on the temperature distribution and fluid distribution of the end region of a 330-MW water-hydrogen-cooled turbogenerator using the transient electromagnetic field and the fluid–solid conjugated heat transfer method. In order to reduce the temperature rise of the machine end region, various methods such as changing the air gap spacer height, the shelter board width, and the electric shield material were studied [15–17]. Wang et al. established a 3D fluid–thermal coupled model of synchronous condenser end region and analyzed the effect of the cooling medium velocity on the temperature of various components of the rotor [18]. Fan et al. established a 3D thermal field model of the hydrogenerator and studied the influence of the structural parameters on the damper winding heat generation of the hydrogenerator. The study found that the damper bar pitch can affect the losses and heat generation of damper bars [19,20]. Carounagarane et al. analyzed the thermal behavior of a large hydrogenerator under continuous overload operation. The result shows that the temperature rise of the generator with Class F insulation is acceptable under 20% continuous overload operation [21]. However, the thermal analysis is carried out by applying the empirical radiative and convective heat transfer coefficients as the boundary conditions in the thermal model. The temperature calculation result of this method heavily depends on the accuracy of the heat transfer coefficients obtained from empirical formulas, which is often challenging in large hydrogenerators with complex structures.

Numerous scholars have conducted research on the cooling and heat dissipation problems of large synchronous machines. However, the research on complex fluid field distribution and thermal field of a large synchronous machine that employs a full air-cooled fanless cooling system is not yet comprehensive. The cooling air in such a ventilation system is driven by the rotation effect of the rotor support and rotor pole. The cooling fan is canceled to significantly reduce the ventilation loss and the machine enjoys the operation economy and stability. Cooling air is forced to move within such a large dimension rotating rotor, which apparently exhibits a complex fluid distribution within the rotor. To gain a better understanding of the heat dissipation effect of such a ventilation system, and develop it to be used in larger capacity hydrogenerator, a comprehensive investigation on heat generation, cooling airflow distribution, and temperature field of the machine with a fully air-cooled fanless cooling system is highly desired. The coupled fluid–thermal analysis is the most accurate and effective approach to study this issue.

This paper takes a fully air-cooled 145 MW (generating)–170 MW (pumping) pumped storage generator motor as an example, and the 2D transient electromagnetic field and 3D fluid–thermal coupled field of the pumped storage generator motor are calculated and analyzed. Firstly, the 2D time-step finite element model of transient electromagnetic field based on the real machine is established, and the eddy current loss of the damping bar is obtained by the finite element method. The additional loss on the rotor pole surface is calculated using an analytical method. Then, the 3D fluid–thermal coupled model of the fully air-cooled pumped storage generator motor is established, and corresponding governing equations and boundary conditions of the solving region are given. The calculated losses are applied as heat generation sources in the 3D fluid–thermal coupled model. Finally, the

flow rule of cooling air and the temperature distribution of the machine rotor are analyzed in detail.

2. Model Establishment and Calculation of Transient Electromagnetic Filed

2.1. Establishment of the 2D Transient Electromagnetic Model

The 2D transient electromagnetic model established in this paper is based on a 145 MW (generating operation)–170 MW (pumping operation) pumped storage generator motor. The geometric dimensions and electromagnetic parameters of the studied machine in this paper are summarized in Table 1.

Table 1. Geometric Dimensions and Electromagnetic Parameters.

Parameter	Generating Operation	Pumping Operation
Power rating (MW)	145	170
Rated voltage (kV)	13.8	13.8
Rated current (kA)	6.9	7.82
Power factor	0.88	0.91
Efficiency (%)	98.4	98.5
Frequency (Hz)		50
Rated speed (r/min)		200
Air gap (mm)		28
Number of stator slot		360
Number of pole pair		15
Stator outer diameter (mm)		8100
Stator inner diameter (mm)		7240
Width of stator slot (mm)		24.5
Height of stator slot (mm)		159.5
Axial length (mm)		2050
Effective stator axial length (mm)		1611
Rotor outer diameter (mm)		7184
Rotor pole inner diameter (mm)		6644
Width of field winding (mm)		90
Thickness of field winding (mm)		8
Turns of field winding		27
Stator core weight (kg)		109,700
Stator copper weight (kg)		18,186
Rotor copper weight (kg)		27,104

The established 2D transient electromagnetic model is shown in Figure 1. The solving region of the built model includes a stator core, armature winding, air-gap, a pair of rotor poles, rotor yoke, field winding, and damping winding. The damping bars of one damping winding are numbered from 1 to 8 along the clockwise direction and electrically connected at the end. According to the machine's rotation direction, one side of the rotor pole is defined as the windward side, while the other is defined as the leeward side. A total of 21,980 triangle elements are used to divide the solving region and the mesh in the air-gap region and damping windings are refined to enhance the calculation precision.

To simplify the analysis, the following assumptions need to be proposed:

1. The electromagnetic model is built as a 2D electromagnetic field and only the axial component of magnetic vector potential A and current density vector J are considered;
2. The leakage flux outside the machine core is ignored.

In the solution region, the governing equations of the transient electromagnetic field and associated boundary conditions can be derived as follows:

$$\begin{cases} \frac{\partial}{\partial x} \left(\frac{1}{\mu} \cdot \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial y} \left(\frac{1}{\mu} \cdot \frac{\partial A_z}{\partial y} \right) = -J_z + \sigma \frac{\partial A_z}{\partial t} \\ A_z|_{AB} = A_z|_{CD} = 0 \\ A_z|_{AD} = A_z|_{BC} \end{cases} \quad (1)$$